

Consider an operator $A: H \rightarrow H^*$ bounded and linear.

We start from $\sigma_{\text{inf}}(A - zI) = \sqrt{\lambda_{\text{inf}}([A - zI]^* [A - zI])}$ ^{This is the adjoint w.r.t the duality $H^* \times H \rightarrow \mathbb{R}$. This is 4th of pseudo spectra}
by Nick [SPECTRA AND PSEUDOSPECTRA EQ 2.10] using instead of the metric induced by the ℓ_2 norm we use the metric of the HILBERT SPACE where I is an identity and consider the generalised SVD of [LAZZARINO - NAKATSUKASA REMARK 3.2].
In particular we will use [COLBROOK EQ. 3.26] to approximate σ_{inf} using

$$\gamma(z, A) = [\min \{ \inf_{f \in D(A)} R_{z,A}^{(1)}(f), \inf_{f \in D(A^*)} R_{z,A}^{(2)}(f) \}]^{1/2}$$

where $R_{z,A}^{(1)}$ is the RAYLEIGH QUOTIENT of the FOLDED OPERATOR, i.e.

$$R_{z,A}^{(1)}(f) = \langle (A - zI)f, (A - zI)f \rangle_H = \langle (A - zI)^* (A - zI)f, f \rangle_H$$

$$R_{z,A}^{(2)}(f) = \langle (A - zI)^* f, (A - zI)f \rangle_H = \langle (A - zI)(A - zI)^* f, f \rangle_H$$

instead of computing the RAYLEIGH QUOTIENT we rely on solving the eigenvalue problems associated (with a SHIFT AND INVERT transformation to target the bottom of the spectrum, see SLEPC MANUAL).

Q1. Is there a primal dual error in our definition (for the impact of primal dual error see work by Patrick and D. Horn, [LAZZARINO - NAKATSUKASA 2015])?

If A^* is defined by $\forall u, v \in H \quad \langle Au, v \rangle_{H^*} =: \langle u, A^*v \rangle_H$ $H \xrightarrow{A} H^* \not\rightarrow H \xrightarrow{A^*} H^*$
Indeed appears that Matt problem is not well posed!

let go down the path of solving the primal dual error? How can we do this?
Well once again look into [LAZZARINO - NAKATSUKASA 2015] where generalised normal equations are presented, i.e. we stick a Riesz map in the folded operator, i.e. we consider $(A - zI)^* R(A - zI): H \rightarrow H^*$ where $R: H^* \rightarrow H$ and is the canonical Riesz map (the solution operator associated with the inner product on H). How do we relate $(A - zI)^* R(A - zI)$ with the original folded operator?

EXAMPLE

When we consider the folded operator by Matt, we study the eigenvalue problem (in variational form):

$$(*) (\bar{\nabla} \times E, \bar{\nabla} \times F) + (\hat{\nabla} \times H, \hat{\nabla} \times G) - (\bar{z} + z) i (\hat{\nabla} \times H, F) + (\bar{z} + z) i (\bar{\nabla} \times E, G) + \bar{z} z (E, F) + \bar{z} z (H, G) = \lambda [(E, F) + (H, G)]$$

$$A: X \times Y \rightarrow (X \times Y)^* \quad (E, H) \in X \times Y$$

$$\langle A \begin{pmatrix} E \\ H \end{pmatrix}, \begin{pmatrix} E \\ H \end{pmatrix} \rangle_{(X \times Y)^*} = -i (\bar{\nabla} \times E, F) + i (\hat{\nabla} \times H, G)_{L^2}$$

H.W. knowing the form of (*) and our A , can you rewrite the variational form as $(A - zH)^* H^{-1} (A - zH)$?

H.W. Going over [LAZZARINO - NAKATSUKASA REMARK 3.2] and [SPECTRA AND PSEUDOSPECTRA EQ 2.10]. Can you show me why the correct of evaluating the PSEUDOSPECTRA is $\sigma_{\text{inf}}([A - zH]^* H^{-1} [A - zH])$?

Be careful I'm being sloppy we would like $H \in \mathcal{L}_0 \cap \mathcal{L}_{\text{div}}(0)$

If we set $X = \widehat{H_{curl}}$ and $Y = \widehat{H_{curl}}$ then the definition of A always make sense because I can always make sense of L^2 pairing but

$A: \widehat{H_{curl}} \times \widehat{H_{curl}} \rightarrow (\widehat{H_{curl}} \times \widehat{H_{curl}})^* \supseteq L^2$ therefore M^{-1} can not be the L^2 RIESTZ MAP but we can define it as:

$$M: \widehat{H_{curl}} \times \widehat{H_{curl}} \rightarrow (\widehat{H_{curl}} \times \widehat{H_{curl}})^*_{(x,y)} \langle N \begin{pmatrix} E \\ H \end{pmatrix}, \begin{pmatrix} F \\ G \end{pmatrix} \rangle_{(x,y)} = (E, F)_{L^2} + (H, G)_{L^2}$$

$$\text{SOL. H.W. } (A - \varepsilon M)^* M^{-1} (A - \varepsilon M) = (A^* M^{-1} A) - \varepsilon \underbrace{M^* M^{-1} A}_{\text{H.W. PROVE}} - \varepsilon A^* M^{-1} M - \varepsilon^* \varepsilon M^* M^{-1} M$$

This map is SELF ADJOINT (H.W. PROVE)

which means that $M^* = M$

$$= (A^* M^{-1} A) - \varepsilon A - \varepsilon A^* + \varepsilon^* \varepsilon M$$

$$\langle (A - \varepsilon M)^* M^{-1} (A - \varepsilon M) \begin{bmatrix} E \\ H \end{bmatrix}, \begin{bmatrix} F \\ G \end{bmatrix} \rangle = + \varepsilon^* \varepsilon [(E, F)_{L^2} + (H, G)_{L^2}]$$

$$+ \varepsilon [i(\bar{\nabla} \times E, F)_{L^2} - i(\hat{\nabla} \times H, G)_{L^2}]$$

$$+ \varepsilon [i(\bar{\nabla} \times E, F)_{L^2} - i(\hat{\nabla} \times H, G)_{L^2}]$$

H.W. Following calculation in

[LAZZARINO - NAKATSUKASA 2015, EXAMPLE 2] $+ (\bar{\nabla} \times E, \bar{\nabla} \times F) + (\hat{\nabla} \times H, \hat{\nabla} \times G)$



H.W. Do the same calculation for the DIRAC FOLDING.

Let's tackle together:

"H.W. Going over [LAZZARINO - NAKATSUKASA REMARK 3.2] and [SPECTRA AND PSEUDO SPECTRA setting. EQ 2.10]. Can you show me why the correct of evaluating the PSEUDO SPECTRA is $\text{sing}([A - \varepsilon M]^* M^{-1} [A - \varepsilon M])$? "

I'll be sloppy here and discuss in finite dim setting.

Notice in particular that similar ideas might also be in section 44 and 51 of [SPECTRA AND PSEUDO SPECTRA] H.W (Please do your bib check :))

If we introduce a new notion of orthogonality, i.e.

DEFINITION Notice that since M^{-1} is SPD it induces a metric on V , so we define an orthogonal operator $U: K \rightarrow V$ wrt to this metric as $\|Uu\| = \|u\|_K$. You can show (H.W.) that is equivalent to asking that $U^* M^{-1} U = I_K$.

[LAZZARINO - NAKATSUKASA REMARK 3.2]

Now we define the generalised SVD decomposition: $A = U \Sigma V^*$ where $A: K \rightarrow V$ $V^*: K \rightarrow V$ $U: V \rightarrow K$ $\Sigma: K \rightarrow K$ and U, V are orthogonal wrt to metric induced by M^{-1} .

→ H.W Show that (via standard SVD proof) the norm of A^{-1} is the smallest singular value of the generalised SVD.

If you do this you have shown that the smallest singular value (wrt M^{-1}) of $B := (A - \varepsilon M)$ is $\|(A - \varepsilon M)^{-1}\|$.

→ H.W. Show (via standard SVD proof) that eigenvalue of $B^* M^{-1} B$ are the sing.

ular value (wrt M^{-1}) of B .

Dictionary Math commits the "crime" of identifying $L^2 \simeq (L^2)^*$ which means that L^2 RIESZ MAP must go be the identity. So if we do not do this identification we have:

$$M_{L^2}: L^2 \rightarrow (L^2)^* \quad \langle M_{L^2} u, v \rangle_{L^2} = (u, v)_{L^2}$$

$$\sigma_{\text{sing}}(A - zI) = \sqrt{\Delta_{\text{inf}}([A - zI]^* [A - zI])} \sim \sigma_{\text{sing}}(A - zM_{L^2}^{-1}) = \sqrt{\Delta_{\text{inf}}([A - zM_{L^2}^{-1}]^*_{L^2} [A - zM_{L^2}^{-1}]_{L^2})}$$

This means that the PSEUDOSPECTRA we are measuring is $\|(A - zM_{L^2}^{-1})\|_{L^2}$.
But for us $A: H \rightarrow H^*$ so to measure the same thing we need an operator $M: H \rightarrow H^*$ that acts precisely like the L^2 RIESZ MAP.

APPENDIX ADJOINTS

DEFINITION Given $T: V \rightarrow K$ we define its Hilbert adjoint as:

$$T^*: K \rightarrow V \quad (Tu, v)_K = (u, T^*v)_V \quad \forall u \in V, v \in K$$

DEFINITION Let $V, K \subset H$ then we can define the Hilbert adjoint with respect to the pivot space H , i.e. $(Tu, v)_H = (u, T^*v)_H \quad \forall u \in V, v \in K$

DEFINITION Let $T: V \rightarrow K$ we define the adjoint with respect to duality action as $\langle Tu, v \rangle_{H^*} = \langle u, T^*v \rangle_H \quad T^*: H \rightarrow H^*$.

DEFINITION The Banach adjoint $T: V \rightarrow K$ is defined by

$$T^*: K^* \rightarrow V^*$$

$$\begin{array}{ccccc} (T^*f)u & \longmapsto & f(Tu) & \langle T^*f, u \rangle_{V^*} & = & \langle f, Tu \rangle_{K^*} & \text{pick now } H=K=V \text{ and} \\ \in & & \in & & & & \\ K^* & & V & & & & \end{array}$$

get the same adjoint as the one above.

N.B We still need to figure out why the correct notion of adjoint is one with respect to the duality pairing.